



2.13 Changing a Tire

1. Purpose

The purpose of the standard operating procedure is to identify the hazards and define the steps involved in changing a tire.

2. Scope

This procedure is intended for use by all Silverpoint employees .

3. Hazard Assessment & Identification

- Physical stress
- Personal Injury
- Equipment Damage
- Slips, Trips & Falls
- Ergonomics
- Lifting, twisting, pushing, and pulling
- Equipment Failure
- Pinch points
- Air pressure
- Ground conditions
- Weather / lighting conditions
- Potential energy

4. EHS Requirements

- Fire retardant coveralls
- Steel toed boots
- Safety glasses
- Hard hat
- Gloves
- Hearing protection
- Lifting techniques

5. References

- Silverpoint Corporate Codes of Practice
- Alberta OH&S
- Silverpoint Task List
- Silverpoint THA 1.1
- Silverpoint SOP 1.32

6. Procedure



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This procedure can be used for vacuum and water truck, trailers, field and shop tire removal and installation

- 6.1. Choose terrain that is level, free of hazards and safely away from traffic, place the transmission in neutral and set the parking brake. Turn on flashers /beacons. Chock the wheels to prevent the unit from moving.
- 6.2. Remove wheel lug nut locks / indicators (take care not to damage the indicators during removal or installation) on tire boss equipped units the 2 lines from the valve stems to the aluminum hub will need to be removed from the hub.
- 6.3. Using a tire iron / breaker bar, break all the lug nuts free. **(DO not use the TORQUE WRENCH to loosen the lug nuts)**
- 6.4. Place a jack in a suitable location under the truck (Axle/frame rail) Place blocking under the jack to prevent it from sinking into the ground. Jack the unit up until a jack stand can be placed under the unit to support the tire off the ground. Slowly lower the unit onto the jack stand until it is supporting the weight. Ensure the stand is level and securely in place. Leave the jack in place as additional support.
- 6.5. Using a tire iron / breaker bar remove the lug nuts (set aside in a safe place)
 - 6.5.1. **On tire boss equipped unit make a note of the position of the boss lines as they will have to be positioned the same upon installation so they will line up.**
- 6.6. Slide a bar under the tire and while lifting on the bar and rocking the tire pull out on the top of the tire it will start to slide off the hub, on tire boss equipped units the inner tire boss line will need to be routed through one of the holes in the outer rim (be cautious not to break the line or valve while removing the outer tire) If the inner tire needs to be removed place the outer tire in a safe place out of the work area so it does not pose a hazard if it were to fall.
- 6.7. Be sure to inspect tires and rims for any damage /wear before reinstalling onto the unit. Look closely around the rim and bolt holes for abnormal wear, cracks, or any other damage. Ensure rim, wheel hub mounting surfaces, lug nuts and wheel studs are clean and free of debris. (use a wire brush if needed)
- 6.8. To reinstall the tires, use the bar to lift the tire onto the hub and rotate to line up studs with the holes in the rim paying attention to the tire boss line location and slide the rim fully onto the studs, before installing the outer rim ensure the tire boss line will line up to the fitting on the hub. Now lift the outer rim onto the hub and rotate it so the inner tire boss line will go through one of the holes in the outer rim (valve stems should be positioned roughly 180 deg apart) ensure both tire boss lines will be able to be hooked up and reinstall lug nuts.
- 6.9. Lug nuts should be tightened in a crisscross pattern until all the nuts are snugged up
- 6.10. Jack the unit up off the jack stand, remove the stand and slowly lower the unit back onto the ground, and remove the jack.
- 6.11. Set up torque wrench, be sure it is orientated properly for clockwise use only, **Do not use counter clockwise setting, this is for left hand threads only**
- 6.12. Torque the lug nuts now
 - 6.12.1. Vacuum and water truck wheels - **500 ft/lbs.**



6.12.2 . Well site trailers

6.12.2.1 . Nut size 13/16 – **115 ft/lbs.**

6.12.2.2 . Nut size 7/8 – **130ft /lbs.**

6.13. Reinstall the lug nut locks/indicators onto the lug nuts and hook the boss lines back up

6.14. Record date and mileage repair was done into the PMI

6.15. Refer to SOP 2.10 for the re-torque procedure

Remember : if you are in doubt about what to do, call your Supervisor